

# Closed Doors, Open Questions: Crime and Policing After Court Closures

*Work in Progress*

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## Abstract

This paper provides the first neighbourhood-level causal evaluation of the mass closure of over half of all magistrate courts in England and Wales. Using a staggered difference-in-differences design applied to novel neighbourhood-level data on court closures, case transfers, and crime, I find that closures have a null average effect on reported crime but reduce police clearance rates by nearly 10%, consistent with offsetting “discouragement” and “deterrence” effects predicted by a simple theoretical model. By contrast, the transfer of caseloads to a remaining court causes property crime to fall and violent crime to rise by roughly 6% each, with comparable spillovers in neighbouring areas. Effects vary systematically with neighbourhood deprivation and with the affected court’s utilisation rate and operating cost. These in turn are partly driven by rising travel costs and a broad decline in police investigative performance, suggesting that cost-driven judicial consolidation can meaningfully undermine crime reporting and deterrence.

**Keywords:** crime, judicial reform, court closures, police performance

**JEL Codes:** I38, J13, J14, K14, K42

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# 1 Introduction

Well-functioning legal institutions are an important determinant of long-run economic performance (Djankov et al., 2003). By shaping incentives against destructive behaviours, the criminal justice system promotes investment in both human and physical capital, and shapes public trust in civil authority (La Porta et al., 1998; Acemoglu and Johnson, 2005). To do so, its provision must be universal and accessible to victims which has proved to be a challenge for public authorities in view of growing costs. In light of these considerations, and growing pressure on public finances, understanding how the organization of judicial institutions impacts their performance is a crucial policy question (Marciano et al., 2019). Despite this, empirical evaluations of existing reforms remain scarce, instead focusing primarily on the impact of procedural changes at the court-level (Bell et al., 2014; Hernández, 2019; Zorro Medina et al., 2020).

This paper studies the causal effect of the mass closure of courts across England and Wales. Following the 2010 General Election, the new government implemented policies to reduce costs and improve efficiency in the public sector. To this effect, the Ministry of Justice ordered the closure of over 50% of all magistrate courts, the first rung in the judicial system. The expected savings were to be partially reinvested to modernise and improve the efficiency of the remaining courts. I develop a simple theoretical framework that encompasses the two main mechanisms with which I interpret my results, and illustrates how the impact of a policy of court closures has *a priori* an ambiguous effect on crime. Closing courts reduces the willingness of victims to report crimes, due to the inconvenience of having to travel greater distances to go to court, or from the perception that police operate less efficiently and are unlikely to deliver justice (Tarling and Morris, 2010; Madero-Hernandez et al., 2022; Lindo et al., 2020; Dench et al., 2025; Boateng, 2018). I call this the discouragement effect. Alternatively, an efficiently operating court system may deter criminal activity by enhancing the threat of punishment (Nagin, 2013; Chalfin and McCrary, 2017; Denti and Di Cataldo, 2026). This deterrence effect could operate in reverse, if, as a result of court closures, the performance of both the police and the judicial system at large falters. Criminals might then commit more offences. The closing of courts might alter the behaviour of criminals, victims, the police and the courts themselves, leading to an ambiguous effect on crime.

Combining data from the House of Commons and the Police Reporting Archive, I exploit the exogenous timing of court closures to identify the effects of two treatments: the court closures themselves and the associated transfers of court cases to a nearby open court. This is done using a difference-in-difference framework with staggered treatment timing for both reported crime rates and police clearance rates, discussed below. Specifically, causal effects are identified separately using the imputation estimator that compares neighbourhoods affected by either a court closure or a court case transfer with those that are not-yet-treated (Borusyak et al., 2024).

Due to their high cost to victims, I focus on violent and property crime. The results reveal an asymmetric effect between court closures and case transfers. The former see largely null effects, while for the latter, reported rates of property crime fall and violent crime rise each by roughly 6% relative to their monthly averages. Meanwhile, police clearance rates, that are used as a proxy for police performance<sup>1</sup>, fall for both types of crime by just under 10% following a closure. The estimates for case transfers are of a smaller magnitude but reflect a similar effect. An analysis of spatial spillover effects reveals that transfers appear to have more spatially diffuse effects on clearance rates,

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<sup>1</sup>See the discussion by Cook (1979) for more details about clearance rates and police performance.

and more concentrated ones for crime rates. The opposite is the case for court closures.

Subsequent heterogeneity analysis by socio-economic characteristics of communities underscores the differences in effects between areas affected by court closures and court case transfer. For court closures, I find that generally the effects are correlated with the degree of deprivation. For example, in the highest quintiles for level of education the reported rate of violent crime drops by over 12%! Similarly, areas in the first quintile of social housing residents (lowest shares) see declines of just over 12%, while those in the upper quintiles (highest shares) see an increase in reported rates of about 7%. The estimates for court case transfers present a complementary picture: neighbourhoods in the upper quintiles for population density, religious fractionalisation and levels of education see the strongest reduction in reported rates of property with some exceeding 20%. This suggests that the effect of the reform on the behaviour of both victims and criminals varies by the socio-economic make-up of the neighbourhood. Analysis for clearance rates reveals once again that the magnitude of effects is closely linked with the affluence of communities.

Next, I consider the differences in effects among neighbourhoods affected by court closures across the pre-closure characteristics of the court. I find that property crime rates tend to fall by 4% when utilisation rates are highest, but rise when courts have larger operating cost (property crime rates rise by 10% and violent crime by about 5%). This suggests that the size and efficiency of courts impact crime reporting through these channels. Plausibly when a larger court closes, it signals that police monitoring of an area has declined reducing the subjective threat of punishment for criminals, while closures of high utilisation courts may instead lead residents to experience longer waiting times, reducing their willingness to report offences.

In this way, my contribution to the literature is primarily empirical, being the first to consider in detail the effect of court closures on both reported crime and the police at large. I enhance our understanding of public policies aimed at reforming the criminal justice system with an eye for cutting costs and optimising efficiency. My work is thus complementary to existing evaluations of changes to judicial capacity either in the UK or abroad (Espinosa et al., 2017; Denti and Di Cataldo, 2026; Luo, 2026). This literature has mostly produced mixed results. In a paper closest to my own, Luo (2026) finds that following court closures in the UK, waiting times increased by about 1.7 days and the prosecutorial charging rate fell by about 1%, leading to higher crime rates. Equally, when analysing the decision in France to close about 20% of all labour courts, Espinosa et al. (2017) find a fall in demand for litigation and a limited increase in court duration. On the other hand, Denti and Di Cataldo (2026) conclude that the policy of court mergers pursued in Italy boosted judicial efficiency and contributed to limiting the occurrence of property crimes.

The rest of the paper is structured as follows. In section 2, I provide background information on Magistrate Courts and the history of judicial reform in the UK. Section 3 describes the various datasets. Next, section 4 presents the theoretical framework used to interpret the results of the paper. Section 5 introduces the difference-in-difference approach used for identification, while in section 6, I discuss the results for court closures and court case transfers, respectively, including heterogeneity and evaluation of the key mechanisms underlying the baseline results. Section 7 concludes and provides a discussion on the policy relevance of the findings.

## 2 Background

Magistrate courts across England and Wales handle over 95% of all criminal cases, dealing with summary offences (minor crimes) and preliminary hearings for more serious cases. These are staffed by trained volunteers and district judges, who can decide guilt or innocence and can issue sentences up to 12 months' custody.<sup>2</sup> There are therefore frequent interactions between the police when investigating a reported crime, and magistrate courts that handle decisions on bail, warrants and punishments.

In 2010, there were 320 working magistrate courts with an average operating cost of just under £300,000 and a utilisation rate of approximately 40%.<sup>3</sup> (in hours) as a proportion of the maximum capacity of the estate. For more information, see [https://consult.justice.gov.uk/digital-communications/proposal-on-the-provision-of-court-and-tribunal-es/user\\_uploads/reform-estates-national-consultation\\_official-sensitive\\_final.pdf](https://consult.justice.gov.uk/digital-communications/proposal-on-the-provision-of-court-and-tribunal-es/user_uploads/reform-estates-national-consultation_official-sensitive_final.pdf). Following the emergence of the coalition government that pursued greater fiscal consolidation, the Ministry of Justice launched several programmes aimed at “modernizing the justice system, reducing costs, and addressing surplus capacity”. The idea was both to make the justice system more accessible by moving meetings online, and to reduce costs of managing court buildings that had been under-used or become dilapidated over time. As a result of the Court Estate Reform Programme (2010-2014) and the Estates Reform Project (2015 - Present) about 50% of all magistrates courts were closed and staffing was cut by nearly one-third.<sup>4</sup>

Since the 2010s, the reform has remained controversial, as no other European country has seen as many closures as the UK. Moreover, by 2020 only 10% of hearings were held online, while the travel distance to the nearest court, on average, more than doubled. Contrary to expectations, waiting times for decisions on cases in the remaining magistrates courts rose from about 4.5 to 6.5 months. Indeed, in 2018, the Public and Commercial Services Union (PCSU)<sup>5</sup> reported to Parliament that:<sup>6</sup>

*Courts and Tribunals Service is creaking under unrelenting pressure caused by years of chronic underfunding and is largely held together by the goodwill of our members.*

At the same time, academic reactions to the programme have been critical of its excessive focus on cutting costs without adequate measures being implemented to support accessibility (Rossner et al., 2021; Goulondris and McLaughlin, 2026). This problem has only grown worse during the Covid-19 pandemic increasing the backlog of cases significantly (Godfrey et al., 2022). As a result, this set of policies has remained controversial, though few have studied the impact of the policy quantitatively.<sup>7</sup> The rest of this article attempts to do just that from the perspective of crime and police clearance rates.

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<sup>2</sup>For more information about the role of magistrates see <https://www.judiciary.uk/about-the-judiciary/who-are-the-judiciary/magistrates/>.

<sup>3</sup>The utilisation rate is defined as the workload

<sup>4</sup>For more details about the 2010-2014 see <https://www.appgov.org/apg/mp/mpdetail/20775>, and for a detailed description of the HMCTS Reform programme launching in 2015 see <https://www.gov.uk/guidance/the-hmcts-reform-programme>.

<sup>5</sup>This is one of the largest trade unions in the UK, and represents those working for the government or other public bodies.

<sup>6</sup>The full quote, as well as other views on problem across courts, can be found at <https://publications.parliament.uk/pa/cm201919/cmselect/cmjust/190/19008.htm>

<sup>7</sup>The only exception is Luo (2026) who finds a negative effect on judicial performance and a subsequent increase in crime, as reflected in survey data.

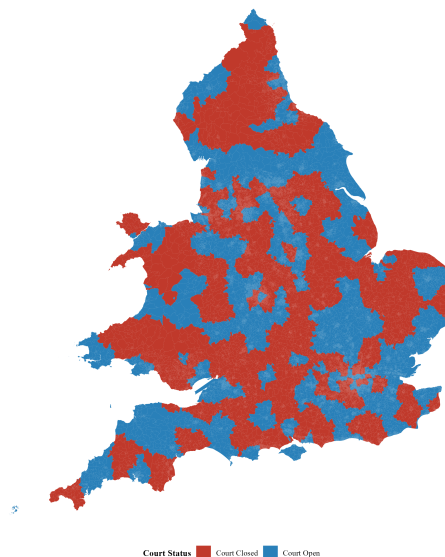
### 3 Data

This article primarily deals with two main types of geographic units: output areas (OAs), and lower layer super output areas (LSOAs). Output areas are the most basic geographic unit used in the compilation of national statistics across England and Wales. An OA contains roughly 300 residents and will be used to form the court jurisdictions discussed below. LSOAs are larger units, typically composed of 5-6 OAs, housing about 1500 residents. In England and Wales in 2011, there were 34,753 such LSOAs. These will be the unit of analysis and are referred to as neighbourhoods.

#### 3.1 Magistrate Courts

Data on magistrate court closures comes from publicly available data compiled from information released to the House of Commons Library.<sup>8</sup> The combined raw data had, at the OA-level, the pre- and post-closure set of court jurisdictions, and information about when each of the closures occurred, the pre-closure court utilisation rate and operating cost. This data was aggregated to the LSOA-level whereby a neighbourhood was considered to be affected by a closure or court case transfer at the earliest treatment date of a constituent output area. For example, if the assigned court of a given output area was closed in January 2013, and this output area was treated earlier than all others belonging to a particular LSOA, then the treatment date for that LSOA becomes January 2013. As noted above, there were 320 magistrate courts in operation in 2010 of which 164 were closed by the end of 2019. Figure 1 illustrates the distribution of closures across England and Wales over the 2010-2019 period.

Figure 1: Magistrate Court Closure



*Notes:* A map illustrating the court jurisdictions and neighbourhoods that were affected by a court closure in the period 2010-2019.

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<sup>8</sup>For more information on magistrate court closures see <https://commonslibrary.parliament.uk/home-affairs/justice/courts/constituency-data-magistrates-court-closures/>.

Closures affect a wide range of communities throughout the country. In addition to basic court-level information, I append data on the change in distance associated with a closure. This is calculated by finding the difference in distance between the centroid of each neighbourhood and the postcode location of the pre- and post-closure magistrate courts. Among those communities affected by a closure, the original distance to the nearest court was about 4 miles, but after closure this rose to over 10 miles.

### 3.2 Local Crime and Clearance Rates

The data for both reported crime and corresponding clearance rate is obtained from police recording archives, available at <https://data.police.uk>.<sup>9</sup> The raw data contains the monthly counts and outcomes for 15 different types of crime at the LSOA-level, and is available from January 2011.<sup>10</sup> I limit the sample to December 2019 to exclude the COVID-19 pandemic period.

For the analysis, I focus on ‘violent’ and ‘property’ crime (as well as their sum total). Not only do these crimes carry the largest social costs, but both violent and property-related offences usually have a specific victim, and are more likely to occur close to or in their neighbourhood of residence.<sup>11</sup> Both conditions are important because one major way in which a court closure may impact crime reporting is through changes in the willingness of the victim to report to the police which may be influenced by changes to the efficiency with which their local court operates. In addition, data on the judicial outcomes of crimes, discussed below, is available for both property and violent crime, allowing me to calculate clearance rates. This is not possible for anti-social behaviour to which no outcome data is attached.

The two types of crime used in the analysis are broader groups obtained from aggregating the raw crime data where property crime is the total number of reported cases of burglary, shoplifting, vehicle crime, bike theft, persons theft and other theft, and violent crime is the total number of violent & sexual offences, robbery, arson and criminal damage, possession of weapons, public order and disorder incidents. This helps to reduce measurement error inherent in using the narrower categories. Changes in Home Office Counting Rules (HOCR), that took effect in April 2018, for violence against persons and crimes relating to stalking and harassment, imply that after that date violent crimes are no longer comparable to the rest of the data. Hence, I discard the affected periods for ‘total crime’ (sum of violent and property crime) and ‘violent crime’.

Clearance rates are calculated also based on the reported crime data as the police attach one of 21 possible outcomes to each offence.<sup>12</sup> There are several possible definitions for

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<sup>9</sup>Data from the British Transport Police has been excluded due to potential measurement error associated with the location of the reported crimes. Data is not available for all LSOAs that fall into the Manchester Police Force Area (PFA) after July 2019 due to misreporting. Note that PFAs are administrative units responsible for the day-to-day running of policing activities and compilation of police data.

<sup>10</sup>Though crime is recorded at the street-level, to preserve anonymity, the publicly available data is recorded at the LSOA-level, a slightly coarser geography. Crime reports are available for 34,750 out of the 34,753 LSOAs that are active during this period.

<sup>11</sup>In Appendix B in Table B.1, I present and discuss the baseline results for reported crime for anti-social behaviour and substance/drug-related crimes.

<sup>12</sup>The outcomes are: “Under investigation”, “Investigation complete; no suspect identified”, “Offender fined”, “Offender given a caution”, “Offender given penalty notice”, “Offender given conditional discharge”, “Offender given community sentence”, “Offender otherwise dealt with”, “Offender given suspended prison sentence”, “Offender sent to prison”, “Defendant found not guilty”, “Court result unavailable”, “Suspect charged as part of another case”, “Local resolution”, “Court case unable to proceed”, “Offender given a drugs possession warning”, “Offender deprived of property”, “Unable to

the clearance rate, but I use a broad one which speaks to how thoroughly the police conduct their investigations: one minus the share of crimes each month for which an investigation was conducted and no suspect was identified (“Investigation complete; no suspect identified”). In calculating the clearance rate, I explicitly do not include any crimes that are still under investigation or where the outcome is not available. This means that my focus lies squarely with those crimes that have gone through the entire investigative process. In a sense, this definition of the clearance rate corresponds to the proportion of all reported offences which were ‘solved’ by the police, and arguably best captures subjective perceptions about police work.<sup>13</sup> Table 1 provides summary statistics for these two crime categories, expressed per 1,000 residents per month at neighbourhood-level, based on the 2011 census population.

Table 1: Reported Crime and Clearance Rate Summary Statistics

| Statistic                       | N         | Mean | St. Dev. | Min  | Max    |
|---------------------------------|-----------|------|----------|------|--------|
| <b>Panel A: Reported Crime</b>  |           |      |          |      |        |
| Total Crime                     | 2,147,694 | 4.98 | 7.01     | 0.00 | 447.52 |
| Violent Crime                   | 2,147,694 | 2.47 | 3.26     | 0.00 | 235.67 |
| Property Crime                  | 2,928,456 | 2.42 | 4.20     | 0.00 | 294.54 |
| <b>Panel B: Clearance Rates</b> |           |      |          |      |        |
| Total Crime                     | 1,689,033 | 0.57 | 0.37     | 0.00 | 1.00   |
| Violent Crime                   | 1,371,299 | 0.67 | 0.40     | 0.00 | 1.00   |
| Property Crime                  | 1,698,302 | 0.55 | 0.37     | 0.00 | 1.00   |

### 3.3 Additional Information

For the heterogeneity analysis, I use a set of control variables defined at the neighbourhood-level from the 2011 Census, available at the NOMIS repository. These characteristics are selected to reflect the socio-economic status of the local area as the personal circumstances of victims may shape their responses to changes in the accessibility of justice. In addition, following the 2008/09 financial crisis, the central government implemented austerity measures to reduce governmental debt. This led to deep cuts to local authority budgets and has been linked to greater crime (Villa, 2024; Giulietti and McConnell, 2020). Hence, I also include a per person estimate of losses (GBP) from austerity cuts at the local authority level, taken from Beatty and Fothergill (2013). Table 2 presents a set of summary statistics for each variable used in the analysis.

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prosecute suspect”, “Offender ordered to pay compensation”, “Defendant sent to Crown Court”, “Formal action is not in the public interest”.

<sup>13</sup>In Appendix B, I consider two other alternative clearance rate definitions that focus on whether the criminal justice system punishes offenders. Hence, following Braakmann et al. (2025), I consider: the proportion of reported crimes where the offender was punished, and the proportion of reported crimes where the perpetrator was identified but was not prosecuted or convicted.

Table 2: Additional Variable Summary Statistics

| Statistic          | N         | Mean   | St. Dev. |
|--------------------|-----------|--------|----------|
| Population Density | 2,936,304 | 35.97  | 43.17    |
| Religious Frac.    | 2,936,304 | 0.47   | 0.10     |
| Education Level    | 2,936,304 | 0.15   | 0.03     |
| Social Rent Share  | 2,936,304 | 0.17   | 0.17     |
| Austerity Loss     | 2,901,528 | 473.91 | 121.74   |

*Notes:* This table provides the summary statistics for the key covariates used in the heterogeneity analysis. For set of detailed definitions see Table A.1 in the Appendix, it provides a brief explanation of each variable.

## 4 Theoretical Framework

In this section, I present a simple model in the style of Becker (1968) illustrating the main mechanisms by which judicial reforms, affecting the number of courts, impact reported crime. Several other studies, such as Chappe and Obidzinski (2014), have studied this theoretical problem in depth, but with a focus on the demand for litigation. In this context, I take a simpler approach to illustrate the mechanisms at work without necessarily specifying all working parts in detail.

The aim of the model is modest: to capture why this effect is a priori ambiguous in that one cannot predict the sign of the relevant treatment effect. Intuitively, this will be due to two countervailing effects: a reduction in deterrence for criminals and a decrease in the willingness to report for residents. The relative strength of these effects determines the overall impact on reported crime and may depend on context, as well as the specifics of the policy.

### 4.1 Criminals

To start with I follow Becker (1968) in defining a market offence function that models the overall number offences that occur over a given period of time:

$$O = O(p, f, z), \quad O_p < 0, \quad O_f < 0 \quad (1)$$

This function implies that offences are determined by the probability of getting convicted ( $p$ ), the severity of the imposed punishment ( $f$ ), and a catch-all term ( $z$ ) that captures all other possible motivations for committing a crime. I impose that, all other things held constant, raising the probability of punishment ( $O_p < 0$ ) or increasing the severity of punishment ( $O_f < 0$ ) reduces the number of offences. Judicial reform, and the closure of courts in particular, likely operates through the former. That is, when fewer resources are directed towards the judicial system, it operates less efficiently as a whole leading to fewer criminals being convicted.

### 4.2 Victims

When crimes are committed, particularly property or violent crimes, victims have to make a decision about whether to report the occurrence of the crime to the police, weighing the costs of reporting with the gains of seeing justice being done. Thus the amount of reported crime will be some fraction of total crime, the reporting rate:

$$r = r(p, d, u), \quad r_p > 0, \quad r_d < 0 \quad (2)$$

Here  $p$  is the probability of conviction,  $d$  is the distance to the nearest court, while  $u$  is once again a catch-all term for all other reasons that may influence a victim's reporting decision. From the victim's point of view, if a criminal is more likely to be convicted of their crime ( $r_p > 0$ ), then it would increase the incentive to report it. On the other hand, court closures would increase the distance needed to go to court if the case is eventually taken for prosecution. For many this is an important cost, especially if the criminal damage was small, leading to a lower reporting rate ( $r_d < 0$ ). Putting everything together, the overall quantity of reported crime is:

$$R = O(p, f, z) \cdot r(p, d, u) \quad (3)$$

In the context of *court closures*, the effect that this reform will have on reported crime can be shown to be ambiguous, as the simple framework above shows that several antagonistic mechanisms are at work.

### 4.3 Courts and Police

Finally, I consider the actions of the court and the police. Let  $N$  denote the number of courts operating in a country or local area. For simplicity, it is assumed that these are distributed evenly across its territory such that, on average, the distance a resident has to travel to their local court is given by:

$$d = d(N), \quad d'(N) < 0. \quad (4)$$

This implies that the average distance is decreasing in the number of open courts. The role of the court is to process a reported crime that has been taken up for prosecution.

Lastly, to keep the mechanisms as simple as possible, denote the probability of conviction as:

$$p = p(\kappa, e, R), \quad p_\kappa > 0, \quad p_e > 0, \quad p_R < 0. \quad (5)$$

where  $\kappa$  is the maximum processing capacity of open courts,  $e$  is police effort and, as above,  $R$  is reported crime. For simplicity, I suppose that the probability of conviction depends positively on judicial efficiency (collectively police effort and court processing capacity) and negatively on the number of reported crimes. I assume that both processing capacity and police effort are functions of the number of courts such that an increase in  $N$  increases both:  $\kappa'(N) > 0$  and  $e'(N) > 0$ .

### 4.4 Equilibrium

I can now state the main results of this model. For simplicity, assume that the terms  $f$ ,  $z$  and  $u$  are independent of  $N$ .

**Proposition 1.** *The first derivative of the number of reported offences by the number of courts is given by:*

$$\frac{dR}{dN} = \frac{(O_p r + O r_p)(p_\kappa \kappa'(N) + p_e e'(N)) + O r_d d'(N)}{1 - (O_p r + O r_p) p_R} \quad (6)$$

*Proof.* See Appendix A.1.

This equation requires interpretation. To make this more straightforward define:

$$\alpha \equiv O_p r + O r_p \quad (7)$$

$$\beta \equiv p_\kappa \kappa' + p_e e' \quad (8)$$

Then the equation above can be written as:

$$\frac{dR}{dN} = \frac{\alpha \beta + O r_d d'(N)}{1 - \alpha p_R} \quad (9)$$

From here, we see that  $\beta > 0$  because  $p_\kappa, p_e, \kappa'(N), e'(N) > 0$ . Similarly,  $O r_d d'(N) > 0$ , as having more courts reduce the distance for victims and raises the reporting rate. However, the sign on  $\alpha$  remains ambiguous. On the one hand, there is a deterrence effect in operation with  $O_p r < 0$ : higher conviction probability means fewer offences. On the other hand, there is a reporting effect whereby  $O r_p > 0$ : higher conviction probability means victims report more. This implies that the sign of the numerator of the overall derivative cannot be determined *a priori*.<sup>14</sup>

Overall, in the case of court closures, this result also applies: one cannot predict whether the reported rate of crime would rise or fall. This depends on the interplay between criminal deterrence, which can be broadly referred to as a “deterrence effect”, and the willingness of victims to come forward and report crimes, a “discouragement effect”. In practice, when courts close, courts that remain in operation and the police would be able to process fewer crimes which both discourages reporting, but also empowers criminal activity. The resulting impact on reported crime depends on the relative strength of these mechanisms, and must be investigated empirically.

## 5 Empirical Methodology

The above framework suggests that judicial reforms, such as those enacted in the UK, have a theoretically ambiguous effect on reported crime, and a generally negative effect on the operation of courts and police investigative activities. Hence, to get a full picture, it is necessary to approach the evaluation empirically.

The base-line analysis studying the impact on reported crime and the police is conducted in a difference-in-difference framework with staggered treatment timing. There are two mutually exclusive and distinct types of treatment: (i) court closure and (ii) the transfer of cases to a specific court that remains open. This means that treatment occurs at the court jurisdiction-level. The two treatments often happen simultaneously, yet affect different sets of communities. For example, when Cardigan magistrates court closes, its cases are immediately taken up by Aberystwyth court, and though both Cardigan and Aberystwyth residents are affected, the causes differ and therefore need to be separately estimated.

To identify the relevant treatment effects, I exploit the exogenous variation in the timing of the court closures and transfers, leveraging ‘not-yet-treated’ units for comparisons. To maintain an adequate sample of control units, the ‘not-yet-treated’ neighbourhoods used are those belonging to a jurisdiction not-yet affected by a closure *or* transfer. I do not use observations from never-treated neighbourhoods, as they are often geographically isolated and would likely be on a different counterfactual trend for both various

<sup>14</sup>This can be extended to the derivative overall because the sign of the denominator is assumed to be positive as a stability condition.

neighbourhood-level characteristics, as well as crime. Formally, I estimate the following specification separately for closure and case transfer:

$$y_{it} = \alpha_i + \eta_t + \beta_{it}D_{it} + \theta_{it} + \varepsilon_{it} \quad (10)$$

where  $y_{it}$  is the outcome (reported crime or the police clearance rate) in neighbourhood  $i$  in month  $t$ ,  $D_{it}$  is the treatment dummy for court closure/case transfer,  $\theta_{it}$  is a vector of interactive fixed effects,  $\alpha_i$  and  $\eta_t$  are the neighbourhood and time fixed effects respectively, while  $\varepsilon_{it}$  is the random error term. In this specification, and given the identification assumptions discussed below,  $\beta_{it}$  is the treatment effect that is allowed to be heterogeneous across both units and time. Importantly, when estimating effects of a closure (cases transfer), I am only using observations in the control group that are unaffected by the transfer of court cases (court closures).<sup>15</sup>

There are two potential challenges to the difference-in-difference design. First, neighbourhoods with different pre-treatment distances to their nearest court may not be directly comparable because residents would respond differently to interventions and changing circumstances. For this reason, in my preferred specification  $\theta_{it}$  includes a set of pre-treatment distance (in deciles) by time fixed effects. Second, austerity shocks, that were not randomly distributed across neighbourhoods, may impact police deployment patterns and may push individuals formerly receiving welfare payments towards criminality, and undermine the parallel trends assumption described below. For this reason, I also include austerity cost (in deciles) by time fixed effects. These are extrapolated from local authority-level estimates, derived by Beatty and Fothergill (2013).

Assuming parallel trends, the average treatment effect on the treated derived from equation (10) could be estimated using a two-way fixed effects (TWFE) model. However, as this is a difference-in-difference setup with staggered timing and heterogeneous treatment effects, TWFE estimator could yield misleading estimates by making “forbidden” comparisons that place ‘already treated’ units in the control group (Goodman-Bacon, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeulle, 2023). Hence, TWFE may not recover the relevant causal effect. To avoid this problem, I use the imputation difference-in-difference estimator developed in Borusyak et al. (2024).<sup>16</sup> In this setting, it uses only the ‘not-yet-treated’ units and periods (those neighbourhood unaffected by closures or case transfers) to impute the missing counterfactuals for the treated neighbourhoods. As the estimator uses all pre-treatment observations for the imputation, it is more efficient than other estimators such as the one developed in Callaway and Sant’Anna (2021). Crucially, it relies on the assumption that:

$$E[\varepsilon_{it}|\alpha_i, \eta_t, \theta_{it}, D_{it}] = E[\varepsilon_{it}|\alpha_i, \eta_t, \theta_{it}] = 0$$

This assumption implies that, conditional on the set of fixed effects, the treatment variable is mean independent of the error term. Alongside the functional form assumption, implicit in equation 10, these imply a simple parallel trends assumption common to most difference-in-difference designs.

Although such assumptions are fundamentally untestable, it is possible to provide evidence in favour of their plausibility. Hence, to evaluate the validity of my approach when running event study analysis, I perform the test suggested in Borusyak et al. (2024). To

<sup>15</sup>I am implicitly assuming that both treatments are “absorbing” states, that is, when a unit is treated it remains treated. Though this holds for court closures, it does not have to for court case transfers. However, only 16 neighbourhoods are affected by multiple transfers, being treated twice in my sample.

<sup>16</sup>For a discussion of this problem, an overview of existing estimators and practical recommendations see the reviews in Baker et al. (2026); De Chaisemartin and d’Haultfoeulle (2023); Roth et al. (2023)

do so, I estimate the following regression by OLS on the set of untreated observations (time periods when neither treatment is active):

$$y_{it} = \eta_t + \alpha_i + \sum_{p=-P}^{-1} \gamma_p 1\{t = \tau_i + p\} + \varepsilon_{it} \quad (11)$$

where  $p$  denotes the number of months before a unit receives treatment in relative time,  $\tau_i$  is a treatment date. The test of parallel pre-trend involves an F-test of the joint significance of all pre-treatment coefficients ( $\gamma$ ). Standard error in the test, as well as the main analysis, are clustered at the pre-closure court jurisdiction-level as that is the level of the treatment. The results of this exercise are separately presented in Figure A.1 in the Appendix where all pre-treatment coefficients are not statistically significant. If plausible, this baseline set of estimates will provide an overall assessment of the impact of court closures and of the accompanying transfer of court cases, on reported crime and clearance rate. However, these estimates will be a composite of several counter-acting mechanisms. The rest of the analysis will focus on picking these apart, and documenting any heterogeneity in treatment effects.

To understand how diffuse these effects are, I evaluate the magnitude of any accompanying spillovers effects. Due to both the size of court jurisdictions that group together many neighbourhoods at a time and the need to restrict my sample to treated areas for identification purposes, estimating spillover effects is challenging.<sup>17</sup> Hence, I re-estimate 10 on the sample of ‘not-yet-treated’ areas that are adjacent to a neighbourhood that is being affected by a closure or case transfer.<sup>18</sup>

Next, I consider the mechanisms driving the baseline results in more detail. First, is there a discernible “discouragement effect” on residents (Lindo et al., 2020). Following the closure of a court, the distance from affected neighbourhoods to their nearest open court increases. This means that going to court takes more time and is costlier (Dench et al., 2025), reducing the willingness of victims to engage with the criminal justice system and reporting crime. Therefore, I check whether the effects of a court closure are greater in neighbourhoods for which either the increase in the cost of travel or distance to the nearest court is larger. Formally, I estimate the following model:

$$y_{it} = \alpha_i + \eta_t + \gamma_{it} \Delta \text{dist}_{it} \theta_{it} + \varepsilon_{it} \quad (12)$$

The interpretation is the same as in equation 10, but where linear measures of distance or cost are used as the treatment variables. In particular, I use the change in distance and the change in the cost of calling a taxi from the centroid of each neighbourhood to the assigned court.<sup>19</sup>

To study treatment effect heterogeneity, I perform heterogeneity analysis across a range of socio-economic characteristics, including population density, level of education, share of residents in social housing, and religious fractionalisation. These characteristics are

<sup>17</sup>Indeed modelling approaches based on estimating effects across a set of concentric rings around any given treated unit is not feasible.

<sup>18</sup>‘Never treated’ units are not additionally included in this analysis because of the difficulty of finding an adequate comparison group for these neighbourhoods.

<sup>19</sup>Taxi cost is used as a proxy for travel costs: data for this is publicly available and can be obtained for any given neighbourhood at <https://www.bettertaxi.com/taxi-fare-calculator/>.

useful proxies for both the degree of urbanisation and deprivation in these neighbourhoods, which could, in turn, shape the relative effect of the policy on the behaviour of residents and criminals. Specifically, I check whether there are patterns in treatment effects across the quintiles of these variables. Moreover, as suggested by the theoretical framework above, the characteristics of the courts themselves may also be an important factor in explaining treatment effect heterogeneity. For this reason, I check to see whether areas where closing courts had higher baseline utilisation rates or operating costs saw greater effects.<sup>20</sup>

Finally, I conduct a set of robustness exercises that can also be interpreted as extensions of my baseline results. As discussed above, court jurisdictions and treatment is based on output areas implying that an LSOA is treated if at least one constituent OA is affected by a closure or court case transfer. In some cases, this may introduce noise into the measurement of the treatment variable. Hence, I estimate the following model:

$$y_{it} = \alpha_i + \eta_t + \beta_{it}S_{it} + \theta_{it} + \varepsilon_{it} \quad (13)$$

The variables are defined in the same way as above, with the exception of  $S_{it}$  which denotes the share of OAs, calculated by using the number treated, in a given LSOA that are affected by a closure or case transfer (as above each type of treatment is estimated separately). This specification may additionally capture the impact of each treatment along the ‘intensive margin’, i.e. how a greater number of treated units impacts local victims and the police. As the treatment variable is no longer binary, treatment effects are captured using the Gardner (2022) two-stage difference-in-difference estimator. This is because it is better able to handle cases where the treatment variable is non-binary. For usual cases of staggered interventions, it yields numerically identical estimates to those of the ‘imputation’ estimator by Borusyak et al. (2024) though differing in their standard errors.

## 6 Results

### 6.1 Baseline

#### 6.1.1 Reported Crime

Table 3 reports the baseline imputation difference-in-difference estimates for the effect of court closures (Panel A) and court case transfers (Panel B) on reported crime. For each crime type, the models progressively contain additional fixed effects, initially including only neighbourhood and time in models (1), (4) and (7), but subsequently adding distance by time, and finally austerity loss by time effects. The model including all fixed effects is preferred because it forces comparisons to be made between neighbourhoods that were a similar initial distance away from each court, and that were similarly affected by austerity policies.

Interestingly, while all estimates for the effect of closures are not statistically significant, case transfers appear to cause larger compositional effects in property and violent crime. In my preferred model, following the transfer of cases, reported rates of property crime fall by just over 6%, while rates of violent crime rise by just under 6% relative to their

<sup>20</sup>Note that information on utilisation and costs was only available for courts that closed, hence this analysis can only be conducted for court closures, and not for the transfer of cases. The basic analysis is supplemented with heterogeneity analysis based on the reason for why a given court was closed. This is discussed in Appendix B.

respective monthly averages.

Given the size of the reform, the asymmetry between closures and case transfers is surprising. Yet, it may be driven by differences in public perceptions of court activity, and yield different institutional responses (Simpson and Pappas, 2024). In particular, courts that have taken on additional cases would appear busier with a greater inflow of defendants, families, witnesses, and bail-related traffic, potentially leading to more police resources being allocated nearby and more police office-hours being spent on court-related duties. The increase in activity can act to deter opportunistic property crime, while greater social activity and friction can generate an increase in minor violent crime incidents.

It is also possible that the effects are partly driven by changes to reporting behaviour. It should be remembered that magistrate courts primarily handle low-value thefts, and hence the benefit of reporting such a crime would be lower, particularly if victims are aware that the likelihood that a suspect is identified has fallen. This is consistent with positive result on violent crime: the willingness to report does not change, while criminals are undeterred by the now dwindling threat of prosecution and unimpressed by the overwhelmed courts that process cases more slowly. Property crime, on the other hand, may be affected more strongly by changes to police efficiency, and hence see a particularly strong reduction in willingness to report.

Table 3: Court Closures and Reported Crime

|                                      | (1)<br>Total      | (2)<br>Total      | (3)<br>Total      | (4)<br>Prop.         | (5)<br>Prop.         | (6)<br>Prop.         | (7)<br>Viol       | (8)<br>Viol       | (9)<br>Viol         |
|--------------------------------------|-------------------|-------------------|-------------------|----------------------|----------------------|----------------------|-------------------|-------------------|---------------------|
| <b>Panel A: Court Closures</b>       |                   |                   |                   |                      |                      |                      |                   |                   |                     |
| Closure                              | 0.096<br>(0.124)  | 0.100<br>(0.127)  | -0.029<br>(0.068) | 0.032<br>(0.036)     | 0.043<br>(0.037)     | 0.028<br>(0.032)     | 0.065<br>(0.103)  | 0.057<br>(0.106)  | -0.057<br>(0.048)   |
| N                                    | 914725            | 914725            | 891074            | 914725               | 914725               | 891074               | 914725            | 914725            | 891074              |
| <b>Panel B: Court Case Transfers</b> |                   |                   |                   |                      |                      |                      |                   |                   |                     |
| Transfers                            | -0.013<br>(0.108) | -0.004<br>(0.102) | -0.008<br>(0.061) | -0.159***<br>(0.038) | -0.156***<br>(0.038) | -0.153***<br>(0.033) | 0.146*<br>(0.087) | 0.152*<br>(0.080) | 0.144***<br>(0.038) |
| N                                    | 932747            | 932747            | 905048            | 932747               | 932747               | 905048               | 932747            | 932747            | 905048              |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Notes: This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of court closures and court case transfers on the reported monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effect. In addition, austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects are added incrementally. The models report the effects on the rates of violent and property crime, and their sum (total crime) defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

Figure 2 presents event study estimates for the effect of court closures and case transfers on each type of reported crime with effects being reported for up to 3 years following each event.<sup>21</sup> It should be noted that all pre-period estimates are both individually and jointly not significant. Hence there is no evidence of pre-trend violations lending some support to the necessary parallel trends assumptions underlying these results, and suggests that there are no anticipation effects at work.

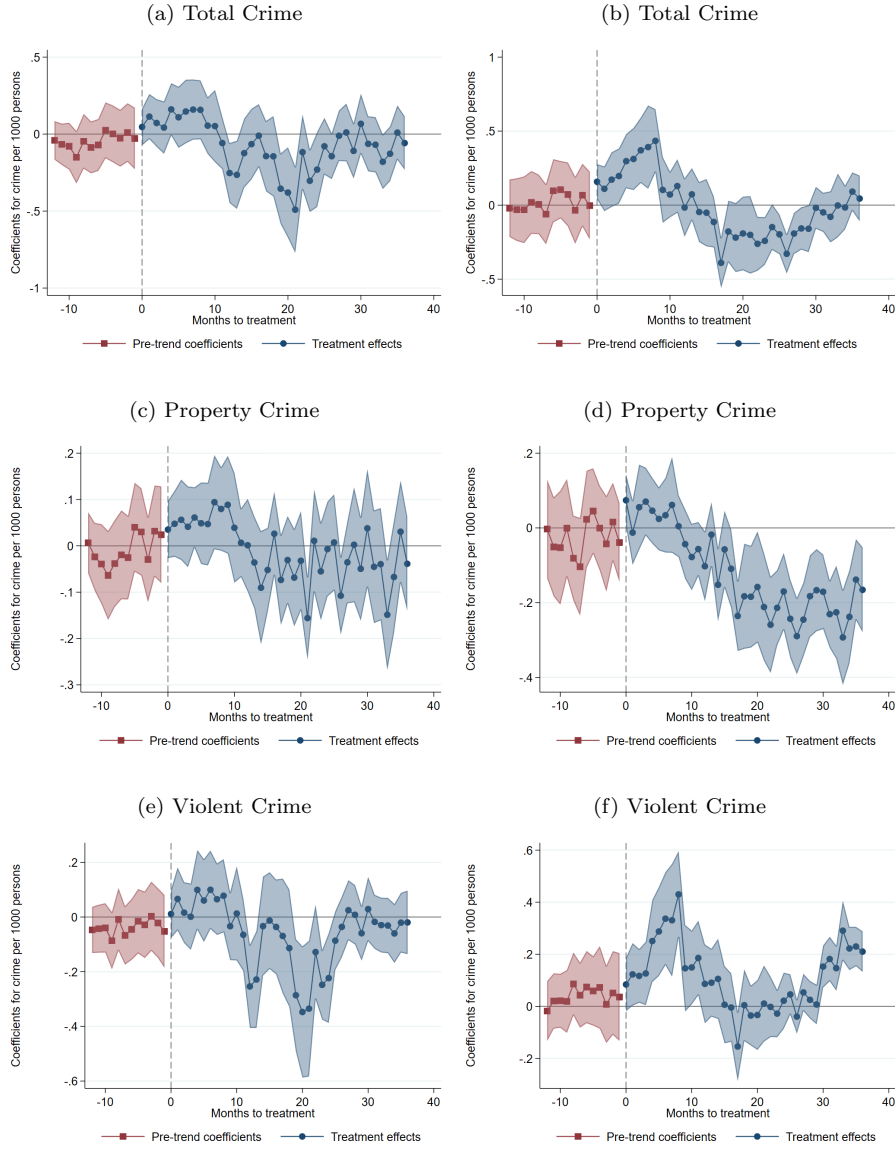
Following a court closure, for each crime type, reported rates decline during the first and second year after a closure, but later recover to their initial level. The initial delay before any decline manifests is broadly consistent with the median length of time from the moment a crime is reported to when a suspect is charged, roughly 6 months.<sup>22</sup> This was driven mostly by reported violent crime that falls initially by 10% relative to its monthly average, and subsequently, by nearly 16.5% by the second year. Property crimes appear to decline in the same period, though the estimates are mostly not significant. The reaction in areas affected by case transfers was more radical. Property crime rates declined around a year after the transfer took place, settling 8% lower than before. This effect appears to be permanent. Violent crime rates responded more variably but generally rose, initially by over 16%, before returning to their average and rebounding again.

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<sup>21</sup>The estimates are normalised relative to the time periods before  $t = -12$ .

<sup>22</sup>This is based on the latest outcome data compiled by the Ministry of Justice and available at <https://www.gov.uk/government/statistics/criminal-court-statistics-quarterly-january-to-march-2025/criminal-court-statistics-quarterly-january-to-march-2025>.

Figure 2: Event Study for Reported Crime



*Notes:* An event study plot for the effect of court closures (left) and case transfers (right) on the reported rates of property and violent crime. A 95% confidence interval is included around each point estimate.

Overall, property crime rates respond similarly in areas with court closures showing declines and indicating the presence of strong negative effects on the willingness of victims to report crimes committed against them. The responses of violent crimes are more asymmetric with closures largely showing a decline in reporting, while areas affected by case transfer instead saw sustained growth. One possible explanation of these estimates is that the relative strength of each type of effect (i.e. “discouragement” and “deterrence”) depends on the local characteristics of each neighbourhood. This possibility is considered through heterogeneity analysis.

### 6.1.2 Clearance Rates

I now consider how court closures and transfers impact the operation of the police. This is measured using the clearance rate of crimes, that is the proportion of all reported offences not dismissed due to no suspect being identified following an investigation. Checking this is important because it underlies two mechanisms used in interpreting the baseline results: criminals being less deterred by the police, and residents being less willing to report. It must be noted that these estimates must be interpreted with care as reporting is not independent of police effectiveness. On the one hand, if victims trust the police, then there is a greater incentive to report less solvable offences, reducing the clearance rates, but ultimately reflecting good performance. On the other hand, if victims only report offences when they expect that the police will solve it, then reported crimes will be skewed towards the simpler cases leading instead to a higher clearance rate with poor police performance.

Table 4 reports the estimates for the effect of court closures and case transfers on police clearance rates. The effects of a court closure for both violent and property crime are negative and statistically significant. In my preferred specifications, the clearance rate for both types of crime declines by just under 10% relative to their monthly means. Meanwhile the effect of transfers tend to be weaker: the estimate for property crime is negative but not statistically significant, while the clearance rate for violent crime falls by just under 8%.

Recall, the estimates above showed that closures, on average, had no effect on reported crime, yet clearance rates still fell. This suggests that court closures and to a lesser extent transfers, that influence the utilisation rates of courts, have had an indirect effect on police activity, presenting a troubling possibility that to counter-act against longer waiting in courts, police are more likely to dismiss cases early on. This supports the idea that both residents and criminals may change their behaviour in response to declines in the efficiency of police investigative activities.

Table 4: Court Closures, Transfers and Clearance Rates

|                                      | (1)<br>Total         | (2)<br>Total         | (3)<br>Total         | (4)<br>Prop.         | (5)<br>Prop.         | (6)<br>Prop.         | (7)<br>Viol          | (8)<br>Viol          | (9)<br>Viol          |
|--------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| <b>Panel A: Court Closures</b>       |                      |                      |                      |                      |                      |                      |                      |                      |                      |
| Closure                              | -0.072***<br>(0.027) | -0.074***<br>(0.018) | -0.054***<br>(0.013) | -0.066***<br>(0.024) | -0.068***<br>(0.017) | -0.049***<br>(0.012) | -0.087***<br>(0.034) | -0.088***<br>(0.023) | -0.063***<br>(0.015) |
| N                                    | 381954               | 381954               | 352989               | 381954               | 381954               | 352989               | 381954               | 381954               | 352989               |
| <b>Panel B: Court Case Transfers</b> |                      |                      |                      |                      |                      |                      |                      |                      |                      |
| Transfer                             | -0.058***<br>(0.029) | -0.062***<br>(0.021) | -0.033***<br>(0.017) | -0.051*<br>(0.027)   | -0.056***<br>(0.020) | -0.026<br>(0.016)    | -0.075***<br>(0.036) | -0.077***<br>(0.026) | -0.048***<br>(0.020) |
| N                                    | 354421               | 354421               | 319478               | 354421               | 354421               | 319478               | 354421               | 354421               | 319478               |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

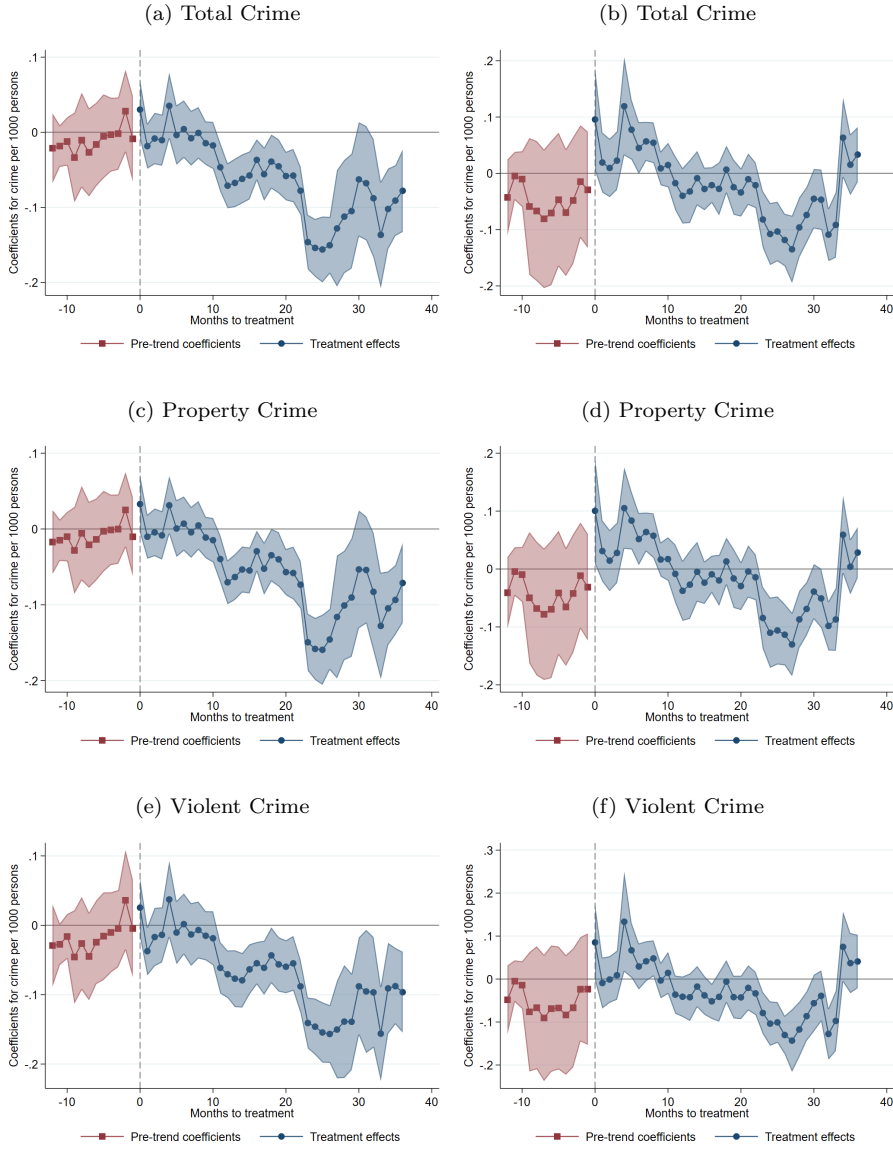
*Notes:* This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of both court closures and of receiving new court cases from closed courts on the monthly crime clearance rate at the LSOA-level. Each model uses time, neighbourhood austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

Figure 3 presents the set of event studies for the effect of courts closures and case transfers on clearance rates over a 3-year period. The first column presents the set of estimates for closures, while the second present those for transfers. It should be noted that in each case the set of pre-trend estimates is again not statistically significant. The general results, apparent from a visual inspection of each plot, is that there is a large and sustained decline in clearance rates for both treatments and all crime types around 2-years following the event, with some evidence of limited recovery soon after.

Clearance rates in areas after a closure fall in two stages. The first occurs between 12-24 months, where declines range from 15-20% relative to the monthly means. Thereafter, there is a second fall lasting about 6-8 months where clearance rates decline up to about a third before undergoing a limited recovery. Hence, though this collapse is partly transitory, it still has the potential to make a significant impact on perceptions of police work. Neighbourhoods impacted by court case transfer do not respond as strongly. There is some evidence even of a short-term increase in clearance rates just after cases are transferred! Yet, as with court closure, 24 months after treatment, there is a sharp decline, particularly in the clearance rate of property crime reaching nearly 20%. However, this fall only lasts about a year before recovery begins and clearance rates return to their pre-treatment values.

Overall, there are several important results that need to be kept in mind. The first is that clearance rates unequivocally decline as a result of both types of treatment. As a result, it is plausible that residents may be incentivised to report fewer crimes, perceiving that police work is less effective, and that criminals may commit more offences as the threat of prosecution and punishment declines. Interestingly, the effect is large and meaningful, but is felt more strongly in neighbourhoods affected by court closures.

Figure 3: Closures, Transfers and Clearance Rates



*Notes:* A set of event study plots for both the effect of court closures (first column) and court case transfers (second column) on the clearance rates of property and violent crime. A 95% confidence interval is included around each point estimate.

## 6.2 Spatial Spillover Effects

Table 5 presents the treatment effect estimates on units whose neighbouring areas have been affected by either a court closure (Panel A) or court case transfer (Panel B). I start by noting that most of the case transfer effects are at best weakly significant, though generally positive. The spatial effects of court closures are more interesting. Again, there is a weak positive effect of about 3% on violent crime, yet the effect on property crime is considerably stronger, amounting to a decline of just under 9%. This has several interesting implications. The first is that if the baseline estimates are plausible in and

of themselves, then this implies that the effect of court case transfers is concentrated in the affected court jurisdiction, while a court closure has wider geographic effects. The second is that in the neighbouring communities property crime falls and violent crime rises, matching the effect that court transfers were found to have above. This suggests that the mechanisms explaining this effect are likely similar.

Table 5: Spatial Spillover Effects - Reported Crime

|                                      | (1)<br>Total     | (2)<br>Total     | (3)<br>Total      | (4)<br>Prop.       | (5)<br>Prop.       | (6)<br>Prop.         | (7)<br>Viol         | (8)<br>Viol         | (9)<br>Viol       |
|--------------------------------------|------------------|------------------|-------------------|--------------------|--------------------|----------------------|---------------------|---------------------|-------------------|
| <b>Panel A: Court Closures</b>       |                  |                  |                   |                    |                    |                      |                     |                     |                   |
| Closure                              | 0.037<br>(0.105) | 0.008<br>(0.099) | -0.130<br>(0.081) | -0.086<br>(0.082)  | -0.141*<br>(0.077) | -0.212***<br>(0.065) | 0.123***<br>(0.045) | 0.149***<br>(0.043) | 0.082*<br>(0.045) |
| N                                    | 634540           | 634540           | 629191            | 634540             | 634540             | 629191               | 634540              | 634540              | 629191            |
| <b>Panel B: Court Case Transfers</b> |                  |                  |                   |                    |                    |                      |                     |                     |                   |
| Transfer                             | 0.091<br>(0.079) | 0.104<br>(0.076) | 0.140*<br>(0.076) | 0.113**<br>(0.046) | 0.058<br>(0.043)   | 0.065<br>(0.049)     | -0.021<br>(0.054)   | 0.046<br>(0.050)    | 0.074*<br>(0.045) |
| N                                    | 646924           | 646924           | 641642            | 646924             | 646924             | 641642               | 646924              | 646924              | 641642            |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Notes: This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of neighbouring court closures and court case transfers on the reported monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effect. In addition, austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects are added incrementally. The models report the effects on the rates of violent and property crime, and their sum (total crime) defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

Table 6 presents the set of effects for court closure (Panel A) and court case transfer (Panel B) on clearance rates. This time, we observe a different pattern from before: all effect of court closures are not significant, while in areas affected by transfers, clearance rates fall sharply by about 20% for violent crime and by just under 10% for property crime. This implies that transfers appear to have more spatially diffuse effects on clearance rates, and more concentrated ones for crime rates.

Table 6: Spatial Spillover Effects - Clearance Rates

|                                      | (1)<br>Total        | (2)<br>Total         | (3)<br>Total         | (4)<br>Prop.       | (5)<br>Prop.         | (6)<br>Prop.         | (7)<br>Viol          | (8)<br>Viol          | (9)<br>Viol          |
|--------------------------------------|---------------------|----------------------|----------------------|--------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| <b>Panel A: Court Closures</b>       |                     |                      |                      |                    |                      |                      |                      |                      |                      |
| Closure                              | 0.010<br>(0.021)    | -0.013<br>(0.019)    | -0.010<br>(0.021)    | 0.014<br>(0.022)   | -0.007<br>(0.020)    | -0.005<br>(0.021)    | 0.014<br>(0.024)     | -0.024<br>(0.020)    | -0.023<br>(0.022)    |
| N                                    | 303147              | 303147               | 300449               | 305747             | 305747               | 302994               | 225618               | 225618               | 223764               |
| <b>Panel B: Court Case Transfers</b> |                     |                      |                      |                    |                      |                      |                      |                      |                      |
| Transfer                             | -0.044**<br>(0.019) | -0.079***<br>(0.017) | -0.066***<br>(0.021) | -0.033*<br>(0.018) | -0.065***<br>(0.016) | -0.053***<br>(0.019) | -0.103***<br>(0.022) | -0.152***<br>(0.015) | -0.137***<br>(0.021) |
| N                                    | 311757              | 311757               | 309083               | 314414             | 314414               | 311685               | 231697               | 231697               | 229865               |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Notes:* This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of neighbouring court closures and court case transfers on the monthly clearance rate at the LSOA-level. Each model uses time and neighbourhood fixed effect. In addition, austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects are added incrementally. The models report the effects on the rates of violent and property crime, and their sum (total crime). Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

Lastly, I check the overall effect of each type of treatment on crime and clearance rate in both directly ‘treated’ units and their neighbours jointly. The results for this exercise are presented in Tables A.2 and A.3 in Appendix A. For reported crime, the results are not impacted greatly: closures have a null effect, while transfers reduce property crime and raise violent crime. For clearance rates, the situation is also similar, though the fall is most pronounced for violent crime.

### 6.3 Discouragement, Distance and Cost

I examine the differential impact that distance/travel cost has on reported crime. This is akin to recasting the baseline difference-in-difference design as one with a continuous treatment variable. Here care must be taken in interpreting the coefficients as the parallel trends assumption introduced above does not allow me to make claims about marginal increases in distance i.e. the causal effect of an increase in distance by one unit. Instead, I merely consider effects relative to distance remaining unchanged.

The results of the two-stage difference-in-difference estimation are given in Table 7 below. For each type of crime, the effect of distance is not statistically significant. However, there is a strong and significant negative effect of travel cost, measured using available taxi quotes, on property crime. Indeed, reported rates fall by nearly 6%. The difference in the estimates can be attributed to travel costs being a more accurate reflection of local infrastructure than merely direct distance. Overall, there seems to be evidence for the role that inconvenience plays in driving the “discouragement effect” in crime reporting.

Table 7: Distance Effects, Taxi Costs and Crime

|       | (1)<br>Total        | (2)<br>Prop.        | (3)<br>Viol        | (4)<br>Total         | (5)<br>Prop.          | (6)<br>Viol         |
|-------|---------------------|---------------------|--------------------|----------------------|-----------------------|---------------------|
| Dist. | -0.0147<br>(0.0255) | -0.0270<br>(0.0193) | 0.0124<br>(0.0129) |                      |                       |                     |
| Taxi  |                     |                     |                    | -0.1532*<br>(0.0896) | -0.1408**<br>(0.0667) | -0.0124<br>(0.0348) |
| N     | 891,065             | 891,065             | 891,065            | 326,118              | 326,118               | 326,118             |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Notes:* This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of court closures on the reported monthly crime rate at the LSOA-level. In these models treatment is a continuous variable corresponding to either (i) the change in distance from before to after a court closure (as in models 1-3), or the increase in the cost of a taxi to the nearest court (as in models 4-6). Each model has time, neighbourhood, austerity loss by time and pre-treatment distance to court by time fixed effects. The models report the effects on the rates of violent and property crime, and their sum (total crime) defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

### 6.4 Heterogeneity Analysis

#### 6.4.1 Reported Crime

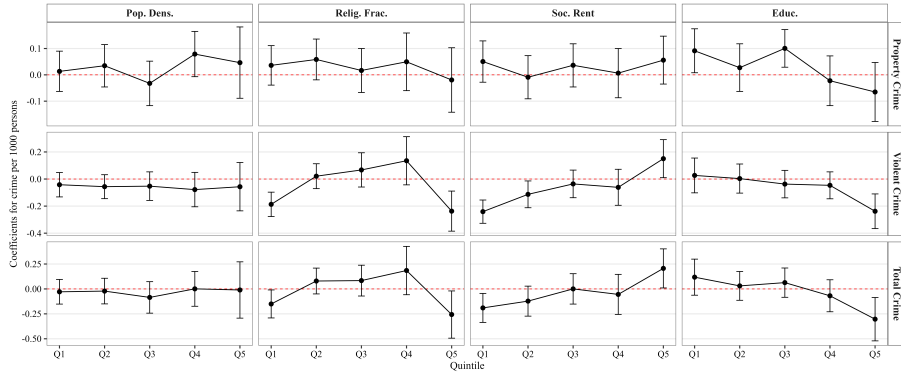
I now consider how treatment effects vary across the socio-economic characteristics of each neighbourhood. Figure 4 presents the heterogeneity analysis for court closures

across the quintiles of four main variables: population density, religious fractionalisation, share of residents living in social housing and level of education.

Rates of reported property crime appear relatively homogenous across all characteristics. One notable exception is the level of education where neighbourhoods at the first and third quintiles experience an increase in reported rates of about 4%, while all other estimates are not statistically significant. Violent crime responds very differently: reported rates fall by over 12% in the highest education quintiles suggesting that more educated, and hence more affluent areas see the largest reduction in their willingness to report offences.

This is more apparent from the estimates using the share of residents living in social housing where areas in the first quintile (lowest shares) see declines of just over 12%, while those in the upper quintiles (highest shares) see an increase in reported rates of about 7%. Similarly, areas with the lowest rates of deprivation and hence those in the lowest quantiles of the crime distribution, see the largest reductions in reporting. In contrast, deprived neighbourhoods where crime is likely more common, see a jump in the number of reported offences.

Figure 4: Heterogeneity Analysis for Court Closures



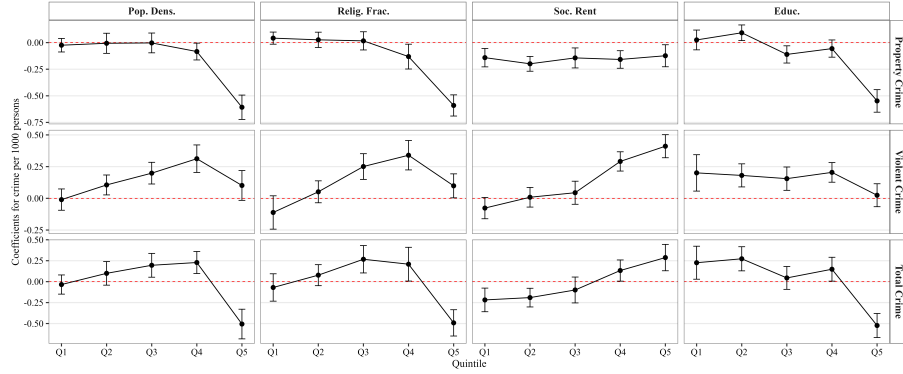
*Notes:* Heterogeneity analysis estimating the effect of court closures on reported crime rates at different quintiles of four socio-economic characteristics of each neighbourhood: population density, religious fractionalisation, share of residents living in social housing and level of education. A 95% confidence interval is included around each point estimate.

Turning now to case transfers, Figure 5 presents the heterogeneity analysis results for the same set of characteristics as above. The patterns here appear to be significantly stronger than for court closures. Starting with property crime, I find that neighbourhoods in the upper quintiles for population density, religious fractionalisation and levels of education see the strongest reduction in reported rates. These effects are often large and economically significant, showing declines that exceed 20%. This seems to be consistent with the previous results: areas that are the most affluent are primarily affected by the “discouragement” mechanism. Although the reason for why the effect of court case transfers cannot be determined for certain, it is likely that local residents are more aware of police-court efficiency in areas where the court is still visibly operating, and hence likewise reduce their willingness to report minor crimes more strongly.

The picture for violent crimes is more nuanced, though retains some of the same basic elements. For example, there is a clear increase in reporting by share of residents living

in social housing with the highest quintiles seeing growth over 15%. Equally, the only insignificant effect for levels of education occurs at the highest quintile. In this way, again it appears as though deprivation is correlated with the magnitude with which violent crime rises following a transfer of cases. However, when split by population density and religious fractionalisation, the effects resemble a left-skewed parabola, though estimates remain positive.

Figure 5: Heterogeneity Analysis for Case Transfer

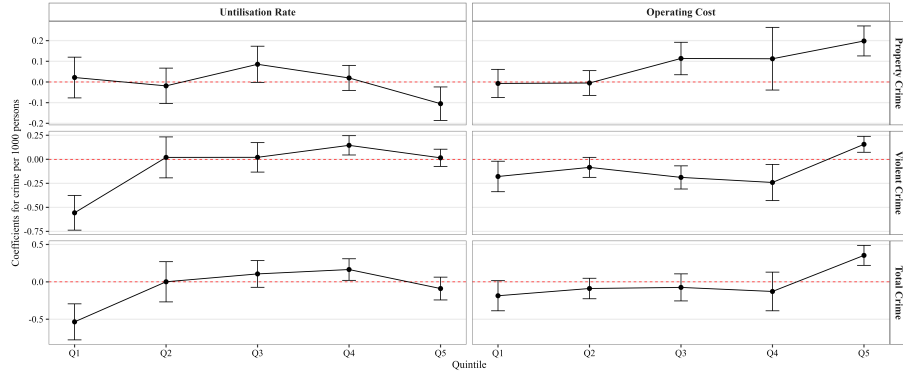


*Notes:* Heterogeneity analysis estimating the effect of court case transfers on reported crime rates at different quintiles of four socio-economic characteristics of each neighbourhood: population density, religious fractionalization, share of residents living in social housing and level of education. A 95% confidence interval is included around each point estimate.

In sum, I find evidence that the effects are correlated with the degree of deprivation in neighbourhoods, with those on the lower end of the score exhibiting larger declines in reported rates for both property and violent crime, while those on the upper end, instead see substantial growth. This is particularly strong in areas directly affected by the transfer of cases, as opposed to those impacted simply by a court closure. This asymmetry is potentially driven by differences in perceptions about police and court efficiency.

In addition to the socio-economic variables, I also consider how the characteristics of closing courts shape their effects on reported crime. Sadly data for courts that did not close is not available, hence this analysis could not be performed for those neighbourhoods. Figure 6 presents the estimates by the quintiles of the pre-closure court utilisation rate and annual operating cost.

Figure 6: Heterogeneity Analysis by Court Characteristics



*Notes:* Heterogeneity analysis estimating the effect of court closures on reported crime rates at different quintiles of court utilisation rates and operating costs. A 95% confidence interval is included around each point estimate.

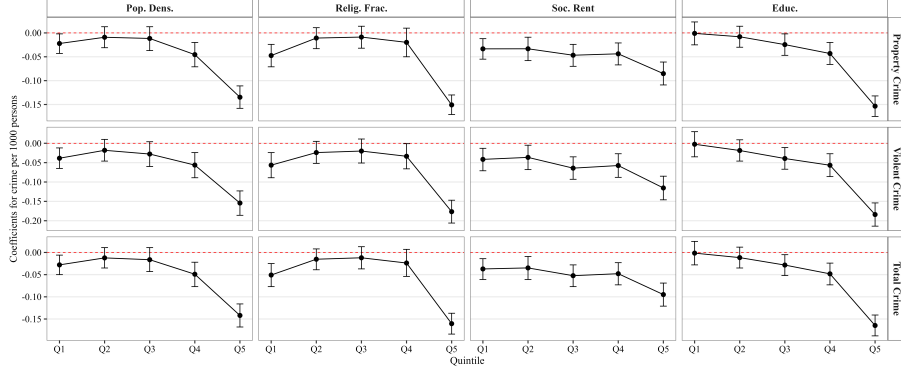
As before there is a contrast between the results for property and violent crime. The former declines by 4% at the highest quintiles of the court utilisation rate, but rises by nearly 10% for the same quintile of annual operating costs. The difference can potentially be explained by the incidence of these effects. A higher utilisation rate is often associated with longer waiting times and potentially pressure on police to reduce the number of prosecuted offences. Hence, closures of courts with a higher utilisation rate may impact residents more strongly as they perceive that the benefit of reporting a given crime has fallen. Meanwhile, courts with larger operating costs are usually those that are visibly larger and more active in the local community. Career criminals would likely be more aware of its activity, and hence its closure would reduce the perceived cost of committing crimes in the area. The reported rate for violent crimes responds differently, instead declining by over 20% for courts with a lower utilisation rate and rising by about 5% for courts in the upper quintiles of the operating cost distribution. This strong negative effect can be partly accounted for by the negative correlation ( $-0.18$ ) between the change in distance and pre-treatment utilisation rate of the courts. That is, often when a court with a lower utilisation rate closed, the distance to the nearest open court was larger leading to stronger “discouragement effects”.

#### 6.4.2 Clearance Rates

Next, I consider whether the impact on clearance rates across various socio-economic characteristics mirrors those of the baseline results for reported crime. Recall, that there I found effects were often correlated with the degree of deprivation in a neighbourhood. Figure 7 illustrates the heterogeneity analysis estimates. The results paint a complementary picture to those of reported crime, though with some major differences. Here the effects on property and violent crime clearance rates is relatively similar in magnitude, and showing the largest decline in the final quintile of each variable. Indeed in high population density areas, for example, the clearance rate of property crime falls by over 30%, while for violent crime by nearly 25%! Overall, the estimates suggest that urban neighbourhoods with greater degrees of religious diversity and higher levels of education see the greatest negative effects. This appears to be more affluent areas, though it must be noted that the impact in neighbourhoods with a large share of residents living in social housing is also significant and negative. These estimates, however, do not contradict

the findings above. Instead, they suggest that in these urban affluent areas, reporting is more strongly affected by perceptions of police work rather than by changes in the behaviour of criminals.

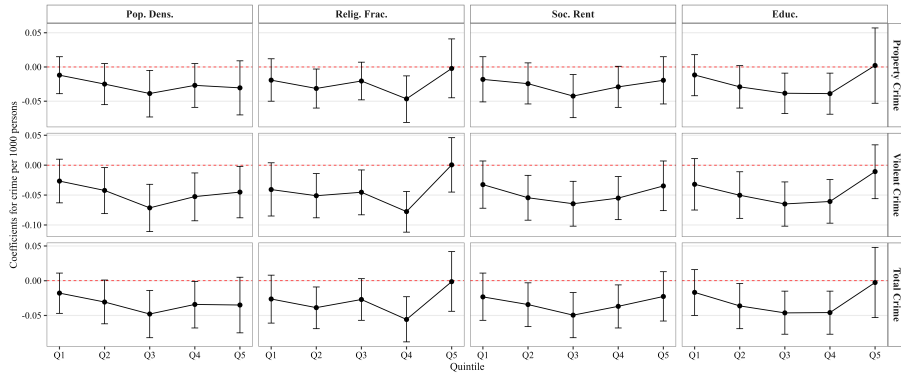
Figure 7: Heterogeneity Analysis



*Notes:* Heterogeneity analysis estimating the effect of court closures on reported crime clearance rates at different quintiles of four socio-economic characteristics of each neighbourhood: population density, religious fractionalization, share of residents living in social housing and level of education. A 95% confidence interval is included around each point estimate.

Next, figure 8 presents the heterogeneity analysis for neighbourhoods impacted by court case transfers. The estimates differ significantly from those above as the largest and most significant effects are concentrated in the middle of the distribution for each covariate, and are generally smaller in magnitude. For example, for the share of residents living in social housing, the largest effect is seen at the third quintile corresponding to a fall in clearance rates of about 9% for violent crime and just under 10% for property crime. A similar pattern seems to occur for population density and level of education.

Figure 8: Heterogeneity Analysis



*Notes:* Heterogeneity analysis estimating the effect of court case transfers on reported crime clearance rates at different quintiles of four socio-economic characteristics of each neighbourhood: population density, religious fractionalization, share of residents living in social housing and level of education. A 95% confidence interval is included around each point estimate.

## 6.5 Robustness Checks and Extensions

Table 8 reports the two-stage difference-in-difference estimates for the effect of closures and transfers, defined using the share of affected output areas, on crime. It must be noted that care should be taken in interpreting these treatment effects as the treatment variable is now continuous (Callaway et al., 2024). To avoid having to implicitly impose a “strong” parallel trend assumption, I will merely consider here the level effects. That is, the difference between having a given share of output areas being treated relative to none being treated. Overall, most estimates align well with the baseline results in Table 3: the effect of closures appears to be insignificant for both types of crime, but transfers cause asymmetric effects on reported crime. In particular, property crime falls significantly by just over 7%. Though the effect on violent crime is large and positive, it is not significant.

Table 8: Closures, Transfers and Share of Output Areas

|           | (1)<br>Total      | (2)<br>Prop.     | (3)<br>Viol       | (4)<br>Total      | (5)<br>Prop.         | (6)<br>Viol      |
|-----------|-------------------|------------------|-------------------|-------------------|----------------------|------------------|
| Closure   | -0.059<br>(0.101) | 0.064<br>(0.049) | -0.123<br>(0.080) |                   |                      |                  |
| Transfers |                   |                  |                   | -0.018<br>(0.134) | -0.164***<br>(0.049) | 0.146<br>(0.106) |
| N         | 909116            | 909116           | 909116            | 927270            | 927270               | 927270           |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Notes:* This table reports the two-stage difference-in-differences estimates for the average treatment effect on the treated of court closures and court case transfers, measured using the share of OAs affected by each type of treatment, on the reported monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effect, as well as austerity loss by time (Aust.  $\times$  Time) and pre-treatment distance (Dist.  $\times$  Time) to court by time fixed effects. The models report the effects on the rates of violent and property crime, and their sum (total crime) defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

## 7 Conclusion

In this paper, I present the first causal estimates of how the closure of more than half of all magistrate courts across England and Wales has impacted reported crime and police activity. To do so I construct a panel from novel data made available by the House of Commons, alongside detailed neighbourhood-level crime and outcome data from the police. Exploiting the exogenous timing of court closures and the associated transfer of court cases, I estimate treatment effects using a staggered difference-in-difference design that makes comparisons between treated neighbourhoods and those not-yet-affected by the policy. Hence, I am in a unique position to study the consequences of such a major reform not only on locals, but also on the criminal-justice system at large.

Surprisingly, I find that court closures have a null direct effect on reported crime rates, but cause falls in the clearance rate of both property and violent crime by just under 10%. This suggests that there are two counteracting mechanisms at work: a “discouragement” effect where residents responds to closures by reducing their willingness to report offences, and a “deterrence” effect as criminals perceive a fall in likelihood of punishment and commit more crimes. As a result, a fall in the clearance rate indicates worsening police efficiency leading to both reductions in reporting and a rise in actual crimes, thus leading to a null effect overall. However, the picture is found to be more nuanced. Neighbourhoods affected by the transfer of court cases, following a closure, experience a rise in reported violent crime by 6%, and an equal decline in reported property crime.

Next, I consider an important underlying channel that explains how the reform may impact reported crime rates: through the effect of distance and travel costs. When a court closes residents have to travel further and perhaps spend more money getting to court and thereby interacting with the criminal justice system. I find that, though distance directly has no effect on reported crime, additional costs, measured using the cost of taking a taxi from the area to the court, reduce reported property crime by as much as 6%.

In addition, I document the heterogeneity of these effects by the socio-economic characteristics of the neighbourhoods. This analysis reveals that the effects are related to the degree of deprivation. Court closures cause the largest negative effects in neighbourhoods with the highest level of education (-12% in violent crime), and lowest share of residents living in social housing (-12% in violent crime). At the same time, the impact of court case transfers follows a similar pattern with areas in the highest quintile for social housing seeing growth in violent crime by over 15%. Yet, there is also evidence that more affluent areas with higher population density and levels of education see the greatest decline in the willingness to report criminal offences. This is also found to hold for clearance rates, particularly of property crime.

Overall, this analysis implies that the policy pursued by the Ministry of Justice with respect to the operation of courts across England and Wales has had far-reaching negative consequences for the criminal justice system and society at large. Not only can this potentially create a large wedge between reported crime and actual crime, but also leads to both an actual and perceived deterioration in the efficacy of police work. This has potentially resulted in both more crimes being committed and a reduction in the willingness of residents to engage with the police increasingly leading to more victims having to bear the cost of crime privately.

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## A Appendix

### A.1 Proofs

*Proof of Proposition 1:*

Recall that reported crime is given by:

$$R = O(p, f, z) \cdot r(p, d, u), \quad (14)$$

where

$$\begin{aligned} p &= p(\kappa(N), e(N), R) & p_\kappa &> 0, & p_e &> 0, & p_R &< 0 \\ d &= d(N) & d' &< 0 \\ \kappa' &> 0 & e' &> 0 \end{aligned}$$

I assume that the catch-all terms  $f$ ,  $z$  and  $u$  are independent of  $N$ . Applying the product and chain rules:

$$\frac{dR}{dN} = (O_p r + O r_p) \frac{dp}{dN} + O r_d d'(N). \quad (15)$$

Since  $p$  is itself a function of  $R$ , the system is simultaneous:

$$\frac{dp}{dN} = p_\kappa \kappa'(N) + p_e e'(N) + p_R \frac{dR}{dN}. \quad (16)$$

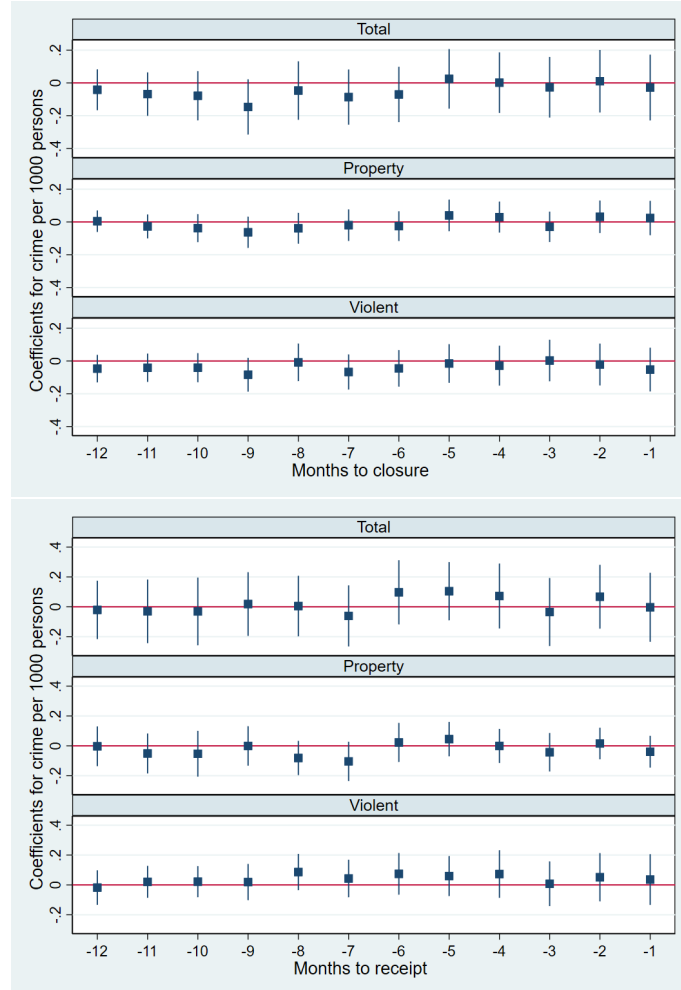
Collecting the  $dR/dN$  terms gives the desired result:

$$\frac{dR}{dN} = \frac{(O_p r + O r_p)(p_\kappa \kappa' + p_e e') + O r_d d'(N)}{1 - (O_p r + O r_p) p_R} \quad (17)$$

## A.2 Figures and Tables

### A.2.1 Figures

Figure A.1: Pre-trend Plots for Court Closures and Case Transfers



*Notes:* A set of pre-trend test plots for court closures and court case transfers. Estimates are normalized using all periods more than 12 months prior to treatment onset. A 95% confidence interval is included around each point estimate.

### A.2.2 Tables

Table A.1: Control Variable Descriptions

| Variable                    | Explanation                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Population Density          | The number of residents of an LSOA per square kilometre.                                                                                                              |
| Religious Fractionalization | Defined using Alesina et al. (2003) as the probability that any two randomly chosen residents belong to two different religions.                                      |
| Education                   | The share of individuals who have achieved at least an L2-level (GCSE) of education.                                                                                  |
| Social Rent Share           | Share of individuals living in socially-rented accommodation.                                                                                                         |
| Austerity Loss              | Estimated per person loss (GBP) from austerity cuts calculated by Beatty and Fothergill (2013). Note that for some local authorities, these values are not available. |

Table A.2: Combined Baseline and Spillovers - Reported Crime

|                                      | (1)<br>Total      | (2)<br>Total     | (3)<br>Total      | (4)<br>Prop.         | (5)<br>Prop.         | (6)<br>Prop.         | (7)<br>Viol      | (8)<br>Viol        | (9)<br>Viol         |
|--------------------------------------|-------------------|------------------|-------------------|----------------------|----------------------|----------------------|------------------|--------------------|---------------------|
| <b>Panel A: Court Closures</b>       |                   |                  |                   |                      |                      |                      |                  |                    |                     |
| Closure                              | 0.076<br>(0.125)  | 0.085<br>(0.122) | -0.071<br>(0.067) | 0.015<br>(0.038)     | 0.021<br>(0.038)     | -0.013<br>(0.034)    | 0.061<br>(0.102) | 0.064<br>(0.101)   | -0.058<br>(0.045)   |
| N                                    | 860192            | 860192           | 837647            | 860192               | 860192               | 837647               | 860192.000       | 860192.000         | 837647              |
| <b>Panel B: Court Case Transfers</b> |                   |                  |                   |                      |                      |                      |                  |                    |                     |
| Transfer                             | -0.011<br>(0.108) | 0.001<br>(0.096) | 0.015<br>(0.061)  | -0.148***<br>(0.038) | -0.150***<br>(0.037) | -0.139***<br>(0.034) | 0.138<br>(0.087) | 0.151**<br>(0.075) | 0.154***<br>(0.037) |
| N                                    | 897311            | 897311           | 870368            | 897311               | 897311               | 870368               | 897311           | 897311             | 870368              |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Notes: This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of both directly treated and neighbouring units resulting from court closures on the reported monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effect. In addition, austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects are added incrementally. The models report the effects on the rates of violent and property crime, and their sum (total crime) defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

Table A.3: Combined Baseline and Spillovers - Clearance Rates

|                                      | (1)<br>Total      | (2)<br>Total      | (3)<br>Total       | (4)<br>Prop.      | (5)<br>Prop.      | (6)<br>Prop.      | (7)<br>Viol        | (8)<br>Viol          | (9)<br>Viol          |
|--------------------------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|--------------------|----------------------|----------------------|
| <b>Panel A: Court Closures</b>       |                   |                   |                    |                   |                   |                   |                    |                      |                      |
| Closure                              | -0.030<br>(0.028) | -0.031<br>(0.020) | -0.021*<br>(0.012) | -0.024<br>(0.025) | -0.025<br>(0.018) | -0.016<br>(0.011) | -0.069*<br>(0.037) | -0.072***<br>(0.027) | -0.050***<br>(0.015) |
| N                                    | 381772            | 381772            | 367145             | 384888            | 384888            | 370194            | 292916             | 292916               | 279837               |
| <b>Panel B: Court Case Transfers</b> |                   |                   |                    |                   |                   |                   |                    |                      |                      |
| Transfer                             | -0.027<br>(0.031) | -0.032<br>(0.021) | -0.012<br>(0.016)  | -0.019<br>(0.029) | -0.024<br>(0.020) | -0.006<br>(0.015) | -0.062<br>(0.039)  | -0.068**<br>(0.027)  | -0.040**<br>(0.019)  |
| N                                    | 383045            | 383045            | 365388             | 386285            | 386285            | 368549            | 291990             | 291990               | 275569               |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Notes: This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of both directly treated and neighbouring units resulting from court closures on the monthly clearance rate at the LSOA-level. Each model uses time and neighbourhood fixed effect. In addition, austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects are added incrementally. The models report the effects on the rates of violent and property crime, and their sum (total crime). Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

## B Additional Results Appendix

### B.1 Anti-Social Behaviour and Substance Crime

Table B.1 presents the results for anti-social behaviour and substance crime (drug-related offences). Columns (1) and (3) are the effects of court closures, while columns (2) and (4) are for court case transfers. The estimates suggest an interesting duality. Although court closures have no direct effect on ASB, court transfer reduce it by about 9.5%. However for substance crime, closures reduce reporting by just under 13%, while court transfer increase it by just over 10%. As noted above, it is challenging to interpret these findings as many of these crimes do not have specific victims and can often be committed further away, perhaps in a different court jurisdiction. Hence, the mechanism by which these effects are generated remains elusive.

Table B.1: Judicial Reform, ASB and Substance Crime

|         | (1)<br>ASB        | (2)<br>ASB           | (3)<br>Substance     | (4)<br>Substance   |
|---------|-------------------|----------------------|----------------------|--------------------|
| Closure | -0.048<br>(0.060) | -0.259***<br>(0.068) | -0.026***<br>(0.007) | 0.021**<br>(0.010) |
| N       | 1567898           | 1707114              | 1567898              | 1707114            |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Notes:* This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of court closures (1 & 3) and receiving new court cases from closed courts (2 & 4) on the reported monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effect, as well as austerity loss by time (Aust. x Time) and pre-treatment distance (Dist. x Time) to court by time fixed effects. The models report the effects on the rates of substance crime and anti-social behaviour defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

### B.2 Clearance Rate Definitions

I use two other definitions of the clearance rate. The first is the proportion of crime, excluding those still under investigation or those where the outcome is unavailable, where the criminal received punishment. The outcomes used to calculate this proportion were: “Offender given suspended prison sentence”, “Offender sent to prison”, “Offender deprived of property”, “Offender fined”, “Offender given a caution”, “Offender given penalty notice”, “Offender given conditional discharge”, “Offender given community sentence”, “Suspect charged as part of another case”, “Local resolution”, “Offender given a drugs possession warning”, and “Offender ordered to pay compensation”. The second tracks the proportion of reported crimes where a suspect was identified but not punished. There I use the following outcomes: “Court case unable to proceed”, “Unable to prosecute suspect” and “Formal action is not in the public interest”. Note that I do not use these definitions in my main analysis because they are driven by several agents in the criminal justice system: police and judges. Instead my broader definition primarily highlights police activity.

Tables B.2 and B.3 present the estimates of the effect of court closures and transfers on the two clearance rates for total, property and violent crimes. The general finding is that

the proportion of criminals receiving punishment increases, and the proportion of crimes where a suspect was identified but not punished has decreased. This holds universally for both property and violent crime. It suggests conditional on getting prosecuted, the probability of being punished has grown. However, in the baseline analysis, we see that a larger share of crimes are being dismissed due to a lack of evidence.

Table B.2: Clearance Rates and Court Closures

|         | (1)<br>Total        | (2)<br>Total         | (3)<br>Prop.        | (4)<br>Prop.         | (5)<br>Viol.        | (6)<br>Viol.         |
|---------|---------------------|----------------------|---------------------|----------------------|---------------------|----------------------|
| Closure | 0.025***<br>(0.007) | -0.084***<br>(0.012) | 0.027***<br>(0.007) | -0.080***<br>(0.011) | 0.022***<br>(0.007) | -0.089***<br>(0.014) |
| N       | 353111              | 353111               | 353111              | 353111               | 353111              | 353111               |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Notes:* This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of court closures on the monthly crime clearance rate at the LSOA-level. Each model uses time, neighbourhood austerity loss by time (Aust.  $\times$  Time) and pre-treatment distance (Dist.  $\times$  Time) to court by time fixed effects. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

Table B.3: Clearance Rates and Case Transfers

|         | (1)<br>Total       | (2)<br>Total         | (3)<br>Prop.       | (4)<br>Prop.         | (5)<br>Viol.       | (6)<br>Viol.         |
|---------|--------------------|----------------------|--------------------|----------------------|--------------------|----------------------|
| Closure | 0.026**<br>(0.011) | -0.062***<br>(0.020) | 0.026**<br>(0.011) | -0.055***<br>(0.018) | 0.026**<br>(0.012) | -0.077***<br>(0.024) |
| N       | 319643             | 319643               | 319643             | 319643               | 319643             | 319643               |

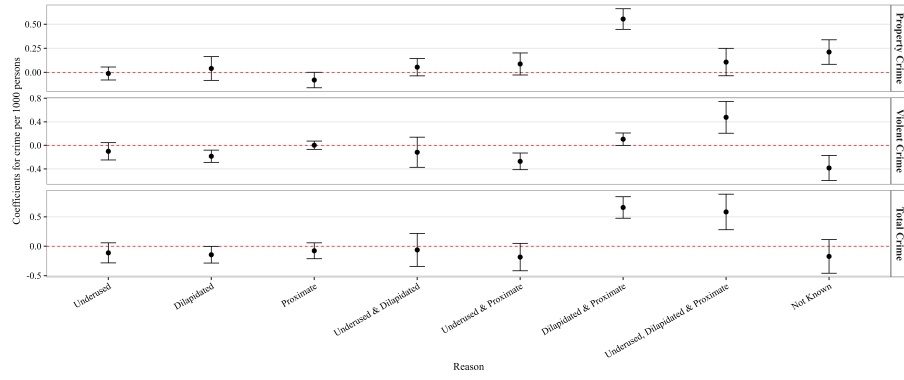
\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Notes:* This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of receiving additional court cases following court closures on the monthly crime clearance rate at the LSOA-level. Each model uses time, neighbourhood austerity loss by time (Aust.  $\times$  Time) and pre-treatment distance (Dist.  $\times$  Time) to court by time fixed effects. Standard errors are reported in parentheses and are clustered at the pre-treatment court jurisdiction level.

### B.3 Reasons for Closure

Here I briefly explore which reasons for closure were associated with the largest effects. Results are presented in Figure B.1. For property crime, this appears to be when the closing court is dilapidated or close to another court that is to remain open, leading to a roughly 10% increase in reporting. A similar sized effect occurs for violent crime, when the closing court is underused, dilapidated and close to another one. These estimates provide some evidence that the mere presence of a court can lead to criminal deterrence even if it is operating suboptimally.

Figure B.1: Heterogeneity Analysis



*Notes:* Heterogeneity analysis estimating the effect of court closures on reported crime rates by reason for closure. A 95% confidence interval is included around each point estimate.